

PHGC: Procedural Heterogeneous Graph Completion for Natural Language Task Verification in Egocentric Videos

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Abstract

*Natural Language-based Egocentric Task Verification (NLETV) aims to equip agents to determine if operation flows of procedural tasks in egocentric videos align with natural language instructions. Describing rules with natural language provides generalizable applications, but also raises cross-modal heterogeneity and hierarchical misalignment challenges. In this paper, we proposed a novel approach termed **Procedural Heterogeneous Graph Completion (PHGC)**, which addresses these challenges with heterogeneous graphs representing the logic in rules and operation flows. Specifically, our PHGC method mainly consists of three key components: (1) *Heterogeneous Graph Construction module* that defines objective states and operation flows as vertices, with temporal and sequential relations as edges. (2) *Cross-Modal Path Finding module* that aligns semantic relations between hierarchical video and text elements. (3) *Discriminative Entity Representation module* excavates hidden entities that integrate general logical relations and discriminative cues to reveal final verification results. Additionally, we further constructed a new dataset called *CSV-NL* comprised of realistic videos. Extensive experiments on the two benchmark datasets covering both digital and physical scenarios, i.e., *EgoTV* and *CSV-NL*, demonstrate that our proposed PHGC establishes state-of-the-art performance across different settings. Our code and dataset are available at <https://github.com/XunCHN/PHGC>.*

1. Introduction

By capturing perspectives of users directly, egocentric video analysis [1–4] allows for real-time task assistance and skill acquisition. In these contexts, the *Natural Language-based Egocentric Task Verification (NLETV)* challenge [5] has recently been proposed to enable agents to comprehend and

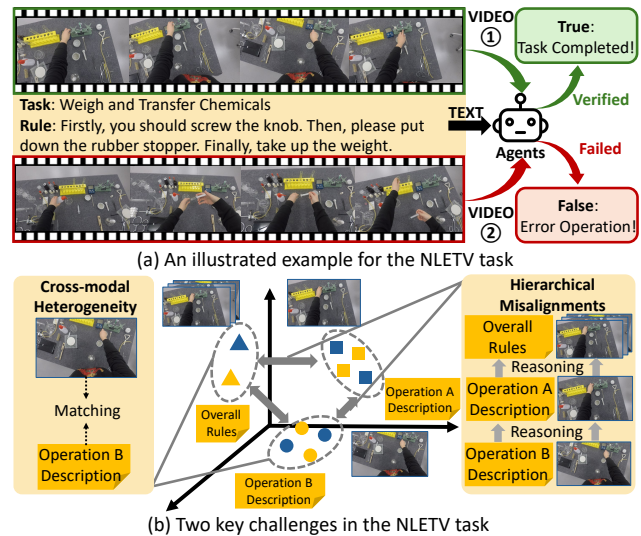


Figure 1. (a) An illustration of the NLETV task. (b) Two challenges: cross-modal heterogeneity and hierarchical misalignment.

verify procedural tasks from egocentric videos based on natural language rules. As illustrated in Fig. 1(a), given language descriptions of a procedural task, the NLETV task aims at determining whether the egocentric videos have accurately completed multi-step tasks, adhering to the temporal order and specified operational criteria.

Compared with previous video-based procedural task verification works [1, 6, 7], the NLETV is more practical and generalizable: describing standard procedures by natural language well fits interactive intuition of humans, while providing better flexibility. However, as depicted in Fig. 1(b) the NLETV task is also highly challenging due to two difficulties: (1) *Cross-modal heterogeneity between text and video data*. Unlike the previous works in video-based task verification [1, 7], the standard procedures in NLETV task are described in natural language instead of the same visual signals as target videos. It requires models to refine rules from the text modality while perceiving key operational cues from egocentric videos. (2) *Hierarchical mis-*

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alignment between procedural logic in text-based rules and the actual operation flows in egocentric videos. The textual descriptions stress the topological relation among different operations, while the videos show how the tasks are conducted comprehensively following linear relations and contain massive irrelevant practical details, making it hard to learn logical relations across different hierarchies.

To address the two issues mentioned above, we proposed a novel NLETV method termed *Procedural Heterogeneous Graph Completion (PHGC)*. It constructs a heterogeneous graph structure to jointly represent textual descriptions and egocentric videos, while progressively conducting graph completion to capture both the logic of rules and the operational flows. As shown in Fig. 2, our proposed PHGC method contains following three key components: (1) Heterogeneous Graph Construction module (HGC), which initializes heterogeneous graphs to depict the textual rules and operation flows in observed egocentric videos. Particularly, the vertices represent hierarchical cues for the verification process, such as object states, partial operation flows or overall task. The edges in the graphs also have diverse attributions including temporal, sequential, or matching relations. (2) Cross-Modal Path Finding module (CMPF), which establishes semantic matching between hierarchical video and text vertices within the heterogeneous graph, enhancing the alignment between visual and linguistic information. (3) Discriminative Entity Representation module (DER), which identifies and highlights the hidden critical entities in heterogeneous graphs to aggregate all discriminative features to deliver an accurate verification result.

Additionally, as the previous works [5] provide the synthetic dataset only, we contribute a realistic NLETV dataset named CSV-NL, which depicts egocentric procedural tasks to supply the evaluation in the physical world. Extensive experiments on both the public EgoTV [5] and our CSV-NL datasets show that our proposed PHGC approach consistently outperforms current state-of-the-art methods.

Overall, our contributions are summarized as follows: (1) We propose a novel NLETV approach, termed PHGC, which effectively addresses cross-modal heterogeneity and hierarchical misalignment and achieves state-of-the-art performance. We also introduce a realistic dataset named CSV-NL to evaluate NLETV models on real-world scenarios. (2) We devise an HGC module that thoroughly represents the NLETV verification process through heterogeneous graphs with hierarchical vertices and diverse relations. (3) We develop the CMPF and DER modules to conduct graph completion, progressively building semantic relations among vertices and extracting discriminative entities.

2. Related Work

Egocentric Video Understanding. The egocentric video understanding focused on analyzing and interpreting video

data captured from a first-person perspective. Early works in egocentric vision have centered on domain adaptation [8–10], multimodal learning [11–14], and video-language pre-training [15, 16] to enhance representations for downstream tasks. With the release of extensive egocentric vision datasets [6, 17–19], more research has shifted toward specific downstream tasks, such as action recognition [20], action anticipation [21, 22], and action segmentation [23]. Recently, researchers have also increasingly focused on developing more sophisticated models capable of handling the egocentric videos as human assistants [5, 24, 25]. Particularly, Rishi *et al.* [5] introduced the language-based procedural task verification into egocentric videos. They aimed to propose egocentric agents capable of tracking and verifying everyday tasks based on natural language specification, inspiring us to develop this work.

Procedural Task Verification. Procedural Task Verification aims to assess whether a sequence of actions or steps in a video has been accurately completed according to a pre-defined procedure. The early works [1–4, 6] mainly focused on video-based task verification, where the standard procedures are described with reference videos. Based on the sequential video datasets [2, 3, 6], a group of works [1, 4, 7, 26, 27] focuses on analyzing the query video concerning the tested videos and estimating their similarities. Although these works above achieve promising results, their verification process is consistently built on video-based demonstrations, which heavily limited their applications. Recently, Rishi *et al.* [5] proposed the Natural Language-based Egocentric Task Verification (NLETV) task, introducing the procedural task verification into egocentric vision and generalizing the video reference to language-based descriptions. Compared with conventional works, the NLETV task is more flexible and fits interactive intuition of humans better, showing more practical meaning especially when it is facilitated with agents.

Heterogeneous Graph Modeling. By representing diverse elements and complicated relations as a graph structure, the heterogeneous graph modeling [28–30] serve as an effective framework for representing and analyzing intricate structures across various domains. It is especially beneficial in multimodal learning tasks [31–34] and video understanding tasks [35–39]. It is a powerful tool for revealing and leveraging these intricate relationships and interactions across diverse modalities [29, 30, 40], as well as handling uncertain data within multimodal learning contexts [13, 41, 42]. Heterogeneous graph modeling has also been adopted in video understanding tasks such as temporal action detection [37], video question answering [36, 43], and video moment retrieval [39, 44, 45]. Inspired by these applications of heterogeneous graph modeling in multimodal learning and video understanding, we also tackle the NLETV task from a graph view and represent the task verification as a proce-

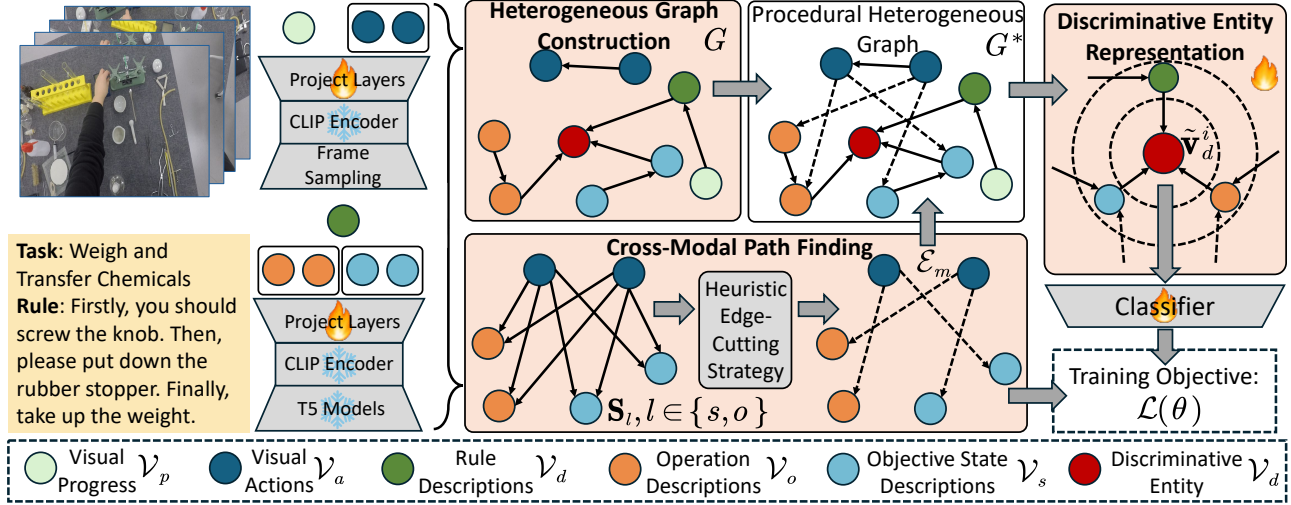


Figure 2. An overview of our proposed PHGC method, which consists of three key components: (1) Heterogeneous Graph Construction that initializes the graph structure; (2) Cross-Modal Path Finding connecting fine-grained visual and textual vertices. (3) Discriminating Entity Representation that aggregates informative cues for final task verification.

dural heterogeneous graph completion process.

3. Our Proposed PHGC Method

3.1. Preliminaries

The NLETV models are given an egocentric video V represented as a sequence of frames or short clips $V = \{v_i | i = 1, 2, \dots, n\}$ and a natural language description D comprised of a sequence of action descriptions $D = \{D_j | j = 1, 2, \dots, m\}$. Here v_i and d_j represent the i -th short video clip and the textual description of the j -th operations in standard task pipeline. The task of NLETV can be formulated to develop a binary classification model $f_\theta(V, D)$ with parameter θ , returning 1 if V aligns with D otherwise 0. Let y represent the labels for (V, D) pairs, it can be formulated as the following optimization problem:

$$\theta^* = \arg \max_{\theta} \log p(f_\theta(V, D) = y | V, D). \quad (1)$$

Particularly, the training data in the NLETV task is only labeled to highlight if a (V, D) pair is matched, without fine-grained temporal annotations for the relevance between operations and action descriptions. This requires models to understand both cross-modal semantical consistency and coherent topological structures for procedural tasks.

3.2. Heterogeneous Graph Construction

We represent the language-based verification process as a heterogeneous graph, where the vertices containing hierarchical semantics from different modalities and edges indicate temporal, topological, and semantically matching relations. In these contexts, the HGC module is devised to initialize a preliminary heterogeneous graph and empower knowledge into the graph-based representation.

Vertices Initialization. As illustrated in Fig. 2, the preliminary heterogeneous graph consists of five types of explicit vertices at different observing levels: (1) Objective state descriptions $\mathcal{V}_s = \{v_s^i | i = 1, 2, \dots, m_s\}$ that depict how the agent interacts with surroundings; (2) Operational descriptions $\mathcal{V}_o = \{v_o^i | i = 1, 2, \dots, m_o\}$ specifying each operation in the procedural task; (3) The rule description $\mathcal{V}_r = \{v_r\}$ for the overall task; (4) Visual actions $\mathcal{V}_a = \{v_a^i | i = 1, 2, \dots, n_a\}$ within the egocentric video; (5) The visual progress $\mathcal{V}_p = \{v_p\}$ presented with the overall video. The former three vertices are extracted from the textual descriptions D hierarchically, while the latter two types are originated from videos V regarding the temporal lengths. Additionally, there is also an implicit vertice, $\mathcal{V}_d = \{v_d\}$, which is the ending vertex performing as a discriminative token for the task verification process.

Edges Initialization. We firstly introduce four types of relations that reveal prior knowledge in the verification process, *i.e.*, temporal relations in visual cues $\mathcal{E}_t = \{e_t^i | i = 1, 2, \dots, n_t\}$, state topological relations $\mathcal{E}_s = \{e_s^i | i = 1, 2, \dots, m'_s\}$ and operational topological relations $\mathcal{E}_o = \{e_o^i | i = 1, 2, \dots, m'_o\}$ provided in language rules, and rule-task matching relation $\mathcal{E}_r = \{e_r\}$ between the visual progress vertex $\mathcal{V}_p$ and the rule description $\mathcal{V}_r$.

Specifically, for the relation originated from text modality, we follow the initial method [5] and adopt the pre-trained language model [46] to generate the objective state and operational descriptions and extract their hidden topological relations. For example, given the rules “*Apple is heated, then cleaned in a sinkbasin and sliced with a knife, then placed in a plate*”, the state descriptions consist of v_s^1 : (apple, hot), v_s^2 : (apple, clean), v_s^3 : (apple, knife, slice),

and v_s^4 : (apple, plate, in), and the state topological relations can be represented as $\mathcal{E}_s = \{e_s^1 : v_s^1 \rightarrow v_s^2, e_s^2 : v_s^1 \rightarrow v_s^3, e_s^3 : v_s^2 \rightarrow v_s^4, e_s^4 : v_s^3 \rightarrow v_s^4\}$. Moreover, the operational descriptions are v_o^1 : (heating the apple), v_o^2 : (cleaning the apple in a sinkbasin), v_o^3 : (slicing apple with a knife), and v_o^4 : (placing the sliced apple in a plate), while the operational topological relations $\mathcal{E}_o$ are similar to $\mathcal{E}_s$.

For the visual modalities, we divide the egocentric to short clips as the visual actions $\mathcal{V}_a$, and connect these clips according to their temporal relations, which are also denoted as $\mathcal{E}_t$. Finally, at the top task-level, we represent the overall language description as $\mathcal{V}_r$ and the complete video as $\mathcal{V}_p$. The visual progress vertex $\mathcal{V}_p$ is connected to the rule description $\mathcal{V}_r$ directly as the matching relations are defined by labels, forming rule-task matching relation the $\mathcal{E}_r$.

Heterogeneous Graph Initialization. Generally, we initialize the heterogeneous graph $G = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V} = \{\mathcal{V}_s \cup \mathcal{V}_o \cup \mathcal{V}_r \cup \mathcal{V}_a \cup \mathcal{V}_p\}$ and $\mathcal{E} = \{\mathcal{E}_t \cup \mathcal{E}_s \cup \mathcal{E}_o \cup \mathcal{E}_r\}$ respectively. Following [5], we employ the frozen CLIP model [47] to represent vertices and further leverage project layers to encode different vertices into $\mathbb{R}^d$. Particularly, for the vertices from video modality, we process each frames with the CLIP image encoder and additionally employ transformer layers to learn contextualized representations.

3.3. Cross-Modal Path Finding

Implicit Semantic Matching Relations. Compared to previous video-based task verification works [1, 6, 7], the NLETV task is more challenging due to the cross-modal heterogeneity: the semantics are implicitly represented in two different modalities at a fine-grained level. This indicates that constructing the potential connections between visual and textual vertices performs an essential role.

To this end, we consider there is the fifth type of edges, *i.e.*, semantic matching relations $\mathcal{E}_m$, connecting fine-grained vertices $\mathcal{V}_s, \mathcal{V}_o$ to the visual actions $\mathcal{V}_a$. We proposed the CMPF module to discover the implicit semantic matching relations $\mathcal{E}_m$ by estimating the similarities between two vertices from visual and textual modalities. Let the $v_l^i, l \in \{s, o\}$ and v_t^j represent the i -th fine-grained textual vertex and the j -th visual vertex respectively, we first estimate the similarity s_l^{ij} between them:

$$s_l^{ij} = \sigma(\mathbf{W}_l v_l^i + \mathbf{W}_t v_t^j + \mathbf{b}), \quad (2)$$

where $\mathbf{W}_l, \mathbf{W}_t \in \mathbb{R}^{d \times 1}$ are learnable weight matrices and $\mathbf{b}$ is bias matrix. $\sigma(\cdot)$ is the sigmoid function. In this way, we calculate a similarity matrix $\mathbf{S}_l = [s_l^{ij}]_{m'_l \times n_t}$ that depicts a sub-graph for $v_l^i, l \in \{s, o\}$ and v_t^j .

Heuristic Edge-Cutting Strategy. With the fully connected similarity matrix $\mathbf{S}_l$, we can preliminarily observe how these vertices are semantically related to each other.

However, a fully connected sub-graph is not suitable for task verification thus we have to conduct appropriate edge-cut to discover the most reliable semantic matching relations $\mathcal{E}_m$. Inspired by the initial work [5], we assume that each state or operational description vertices is connected to one key visual action vertex at most. Based on this assumption, we introduce a heuristic edge-cutting strategy through Dynamic Programming, to select the most appropriate edges making the total similarity the highest. Specifically, let $\mathbf{S}_l$ denote the similarity matrix predicted by models and $\Phi = [\phi_{ij}]_{m'_l \times n_t}$ represent the highest total similarity matrix. ϕ_{ij} denote the the highest total similarity when the $\mathcal{V}_l$ and $\mathcal{V}_t$ containing i and j vertices respectively, $i < m'_l, j < n_t$. The edge-cutting process can be formulated as the following algorithm 1. In our CMPF module, the edge-cutting strategy above will regard the learned similarity matrix as reliable and further build connections between selected vertices heuristically.

Algorithm 1 Heuristic Edge-Cutting Strategy in CMPF.

```

Calculate Similarity Matrix:  $\mathbf{S}_l = [s_l^{ij}]_{m'_l \times n_t}$ 
Define Total Similarity Matrix:  $\Phi = [\phi_{ij}]_{m'_l \times n_t}$ 
Initial edge Set:  $\mathcal{E}_m = \emptyset$ 
%Boundary Conditions
For ( $i = m'_l - 1, j = n_t - 1; i > 0, j > 0; i--, j--$ )
     $\Phi[i, n_t] \leftarrow \mathbf{S}_l[i, n_t], \Phi[m'_l, j] \leftarrow -Inf$ 
%Dynamic Programming
For ( $i = m'_l - 1, j = n_t - 1; i > 0, j > 0; i--, j--$ )
     $\Phi[i, j] \leftarrow \max(\mathbf{S}_l[i, j] + \Phi[i + 1, j + 1], \Phi[i, j + 1])$ 
%Cutting Edges
For ( $i = 1, j = 1; i \leq m'_l; i++$ )
    If  $\Phi[i, j] > \Phi[i, j + 1]$ 
        Then  $\mathcal{E}_m \leftarrow \mathcal{E}_m \cup \{e_m^{ij} : v_t^j \rightarrow v_l^i\}, j++$ 
Output  $\mathcal{E}_m$ 

```

3.4. Discriminative Entity Representation

Procedural Heterogeneous Graph. Due to the fact that hierarchical matching cues and detailed topological and temporal information are not integrated, we consider leveraging task-rule vertices as discriminative entities to bind the multimodal instances and aggregate hierarchical information. Discriminative entities are the ending vertex of the overall graph, which have zero out-degrees. Concretely, the discriminative entities are connected to the tail of the state and operational topological vertices to perceive the general procedural information in our digraph, as well as the rule description and visual progress vertices to integrating the overall cross-modal matching cues. We denote these edges of discriminative entity as $\mathcal{E}_d$. Based on the newly found edges, *i.e.*, $\mathcal{E}_m$ and $\mathcal{E}_d$, we update the preliminary heterogeneous graph into procedural heterogeneous graph by integrating these cross-modal semantic match-

ing edges and initialized topological and temporal edges: $G^* = \{\mathcal{V}, \mathcal{E} \cup \mathcal{E}_m \cup \mathcal{E}_d\}$.

Discriminative Knowledge Aggregation. With the procedural heterogeneous graph, we can represent the overall language-based task verification process thoroughly. In the next, we conduct message passing within it to learn the representative features for classification. Specifically, we employ the graph attention mechanism [48] to aggregate the most discriminative knowledge in the procedural heterogeneous graph. Let v_*^i and $\mathcal{E}_*^i$ be the i -th vertex in the procedural heterogeneous graph G^* and all edges directed to it, we denote the message passing process as follows:

$$\begin{aligned} \tilde{v}_*^i &= v_*^i + \sigma \left(\sum_{e_*^{ij} \in \mathcal{E}_*^i} \alpha^{ij} \mathbf{W}_f v_*^j \right), \text{ where} \\ \alpha^{ij} &= \frac{\exp(\sigma(f_{GA}([\mathbf{W}_f v_*^i; \mathbf{W}_f v_*^j])))}{\sum_{e_*^{ki} \in \mathcal{E}_*^i} \exp(\sigma(f_{GA}([\mathbf{W}_f v_*^i; \mathbf{W}_f v_*^k])))}. \end{aligned} \quad (3)$$

Here e_*^{ij} represent the edge $v_*^j \rightarrow v_*^i$ in the procedural heterogeneous graph. $\mathbf{W}_f$, $\sigma(\cdot)$, and f_{GA} are learnable weight matrix, activate function, and linear layers for graph attention respectively. $\tilde{v}_*^i$ is the updated vertex. In this way, we update all vertices in the procedural heterogeneous graph G by leading the message passing across adjacents through the directed edges. Finally, we extract the ending vertex, *i.e.*, discriminative entities which can perceive global information, to represent the aligning strengths of the given pairs.

Training Objectives. In Eq. 1, NLETV task aims to maximize the post probability of binary classification. In our PHGC method, the optimization is conducted at both fine- and coarse-grained level due to the hierarchical semantics presented in our procedural heterogeneous graph. Hence, we update the optimization goal of our PHGC method as:

$$\begin{aligned} \theta^* &= \arg \max_{\theta} \left[\log p(f_{\theta}(V, D) = y \mid V, D) \right. \\ &\quad \left. + \frac{1}{|\mathcal{E}_m|} \sum_{e_m^{ij} \in \mathcal{E}_m} \log p(s_l^{ij} = y \mid G^*) \right]. \end{aligned} \quad (4)$$

where f_{θ} denote our PHGC model, while θ and y denote the learnable parameters and labels for (V, D) respectively. e_m^{ij} represents the directed semantic matching edge ($v_a^j \rightarrow v_l^i$), $l \in \{s, o\}$ generated by the CMPF module. The second term in this optimization aims to maximize the similarity of selected matching relations based on the max likelihood estimation. Overall, we adopt the negative log-likelihood in our method to train our PHGC approach, which could be formulated as follows:

$$\begin{aligned} \mathcal{L}(\theta) &= -\mathbb{E}_{G^* \sim \mathcal{D}} [y \log f_{\theta}(V_r) + (1 - y) \log (1 - f_{\theta}(V_r))] \\ &\quad - \mathbb{E}_{G^* \sim \mathcal{D}} \mathbb{E}_{e_m^{ij} \sim \mathcal{E}_m} \left[y \log s_l^{ij} + (1 - y) \log (1 - s_l^{ij}) \right]. \end{aligned} \quad (5)$$

4. Experiment

4.1. Evaluation on Synthetic and Realistic Scenarios

Previous EgoTV Dataset. Existing procedural task verification works focus on video-based verification [1, 4, 7] and most of them are conducted on the datasets of the third-person perspectives. Rishi *et al.* proposed the initial benchmark dataset EgoTV [5] for the NLETV challenge, this data is all collected from the digital space. It is the only dataset currently for the NLETV challenge, which is collected from digital space. The synthetic data in this benchmark dataset covers 82 task categories, and is split into 10726, 1080, 700, 2164, and 676 for training, novel task, novel step, novel scene, and abstraction respectively.

Our CSV-NL Dataset. The evaluation of NLETV models on realistic videos from physical space has not been explored yet. To this end, we further construct a novel benchmark, termed CSV-NL, based on the existing public dataset CSV [6]. The proposed CSV-NL dataset contains more than 15496 pairwise instances, covering 14 task categories, and each task is completed with 5 operational sequences. Additionally, every operational sequence is conducted by 20 operators at least. Each egocentric video in the CSV-NL dataset lasts 21 seconds, consisting of 9 visual actions while accompanying language rules with 46 words on average.

Following EgoTV [5], we define three different evaluation scenarios: (1) *Novel Task*: The models will be trained on seen task categories but tested on unseen task categories. (2) *Novel Step*: The tested tasks are completed with different operational flows, such as shuffled or omitted operations. (3) *Novel Operators*: The tested tasks are completed with the same operational flows, but are conducted by different operators. The three scenarios are designed to evaluate respectively the generalization of models on semantic understanding, cross-modal alignment, and robustness towards irrelevant perturbations such as habits of operators. We split it into 10280, 3360, 896, and 960 for training, novel task, novel step, and novel operators. *More details are available in our open-source repository.*

4.2. Implementation Details

We employ pre-trained backbones to extract visual and textual features and use these features in an offline manner. For fairness, we follow [5] to adopt the pre-trained CLIP model [47] as our backbones. For the video modality, we split each video into a group of clips with 20 frames and used their average embeddings as the visual action vertices. As for the text modality, we first employ the pre-trained T5 models used in [5] to extract the objective state and operational topological vertices, then encode them as well as the overall descriptions with CLIP models. All features are projected into a common space with 512 dimensions. Following [5], we also adopt the F1 Score as our metrics.

Method	Visual Backbone	Text Backbone	Multimodal LLM	Novel Tasks	Novel Steps	Novel Scenes	Abstraction	Average
Oracle Model	CLIP	CLIP	-	95.0	96.7	97.6	97.2	96.6
PandaGPT [49]	ImageBind	Vicuna	✓	17.3	5.0	20.9	35.5	19.7
GoldFish [50]	CLIP	LLaMA	✓	33.6	46.6	23.7	37.6	35.4
Otter [51]	CLIP	LLaMA	✓	53.4	65.4	59.7	61.5	60.0
VIOLIN [52]	ResNet	BERT	-	47.4	80.4	85.6	42.5	64.0
MIL-NCE [53]	S3D	Word2vec	-	30.5	69.6	73.5	24.3	49.5
CLIP [47]	Tx	Tx	-	43.9	66.5	72.2	13.6	49.1
CoCa [54]	Tx	Tx	-	51.5	71.6	71.9	43.5	59.6
VideoCLIP [55]	Tx	Tx	-	29.3	67.6	77.9	25.6	50.1
CLIP4Clip [56]	CLIP	CLIP	-	56.2	73.2	74.6	17.5	55.4
NSG [5]	CLIP	CLIP	-	90.0	64.7	84.9	80.4	80.0
PHGC (Ours)	CLIP	CLIP	-	91.7	82.4	87.1	88.0	87.4

Table 1. Comparisons of model performance on the EgoTV benchmark dataset regarding F1-score. Note that *Tx* indicates the pre-training multimodal foundation models based on Transformer architecture. The Multimodal LLMs are highlighted in gray. *Oracle Model* is the fully-supervised models trained with manual temporal annotations for each visual actions

4.3. Overall Comparisons

Following [5], we compare our PHGC method with the existing state-of-the-art NLETV approach NSG [5] and other baseline methods including VIOLIN [52], MIL-NCE [53], CLIP4Clip [56], and also three baselines which perform as the foundation models, *i.e.*, CLIP [47], CoCa [54], and VideoCLIP [55]. Furthermore, we also evaluate three Multimodal LLMs (MLLMs) designed for video understanding, *i.e.*, PandaGPT [49], Otter [51], and GoldFish [50]. The experimental results on the EgoTV and CSV-NL datasets are illustrated in Table 1 and Table 2 respectively.

Model	Novel Tasks	Novel Steps	Novel Operators	Average
PandaGPT [49]	57.64	53.36	61.23	57.41
GoldFish [50]	56.84	58.42	61.50	58.92
Otter [51]	56.96	42.86	47.14	48.99
MIL-NCE [53]	21.48	37.79	33.34	30.87
CLIP4Clip [56]	36.98	51.29	46.48	44.92
VIOLIN [52]	65.25	48.99	52.06	55.43
NSG [5]	66.85	60.35	55.91	61.04
PHGC (Ours)	68.64	67.59	69.25	68.49

Table 2. Comparisons of model performance on the CSV-NL benchmark dataset regarding F1-score. Note that the Multimodal LLMs are highlighted in gray. Other compared baseline methods adopt CLIP [47] for visual and textual backbones.

By comparing our the performance of our approach with other models, we list following observations: (1) Our proposed PHGC method consistently outperforms all other models across both datasets. These results demonstrate the effectiveness of our proposed PHGC method in tackling the NLETV task, and also prove that leveraging heterogeneous graph to represent the verification process is reasonable. (2) By comparing our method with these LLM-based methods

[49–51], our method achieves remarkable improvements across diverse metrics. It indicates that multimodal reasoning in egocentric scenarios is still challenging for MLLMs, which has also been proved in [25]. Additionally, compared with the Oracle Model [5] that are built on manual temporal annotations for each actions, our method achieves the closest performance, which furtherly demonstrate the superiority of our proposed PHGC method.

However, we also note that our model performs less significantly on the realistic benchmark datasets, while the LLM-based counterparts maintains their performance. We speculate this is because the realistic videos in CSV-NL introduce more variability and complexity compared to synthetic data in terms of scene transitions, visual noise, and linguistic ambiguities. The MLLMs, which are pre-trained on large, diverse multimodal corpora containing massive real-world data, may exhibit stronger robustness to the diverse and dynamic nature of real-world data. Nevertheless, it worths noting that our proposed PHGC approach still outperforms these methods and achieves state-of-the-art, demonstrating the effectiveness of introducing heterogeneous graph completion process into the NLETV task.

4.4. Further Study

Ablation Study. Here we use abbreviation for clarity: Novel Tasks (NT), Novel Steps (NS), Novel SCenes (NSC), ABSTraction (ABS), Novel Operators (NO), and Average (AVG). The experimental results are reported in Table 3.

According to the experimental results, we can observe that: (1) Our proposed HGC module is essential for the overall model. By removing the HGC module results in a significant performance drop across all metrics. This highlights the importance of HGC in capturing the underlying logic of task and operation flows. It also demonstrates that considering the verification process as a directed acyclic

EgoTV						
HGC	CMPF	DER	NT	NS	NSC	ABS
-	-	-	64.1	50.7	68.4	61.1
✓	-	-	89.5	58.2	72.7	84.7
✓	✓	-	88.9	62.6	78.8	84.6
✓	-	✓	89.7	63.4	77.6	87.5
✓	✓	✓	91.7	82.4	87.1	88.0

CSV-NL						
HGC	CMPF	DER	NT	NS	NO	AVG
-	-	-	56.1	52.6	55.6	54.8
✓	-	-	60.8	60.3	58.9	60.0
✓	✓	-	63.9	57.0	58.6	59.8
✓	-	✓	63.6	60.3	64.6	62.8
✓	✓	✓	67.3	67.6	69.3	68.1

Table 3. Ablation studies regarding key components in our proposed PHGC method on EgoTV and CSV-NL datasets.

graph is reasonable and effective for NLETV task. (2) Enabling CMPF module significantly improves performance on Novel Steps and Novel Scenes, underscoring its importance in aligning hierarchical video and text elements. This is due to these scenarios requires better generalization and robustness to fine-grained semantic cues, which is exactly addressed by CMPF module. (3) The presence of the DER module leads to significant improvements in Novel Tasks and Abstraction, showcasing its ability to uncover hidden entities and refine task verification from a general view. The reason is the DER module effectively conducts message passing across the procedural heterogeneous graph, and aggregate discriminative cues into the ending entities.

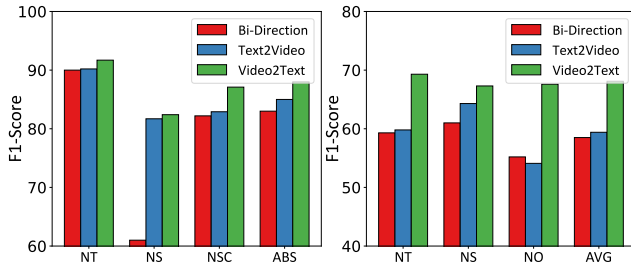


Figure 3. Analysis on the directions of the semantic matching relations $\mathcal{E}_m$ on the EgoTV (Left) and CSV-NL (Right) dataset.

Analysis on Cross-Modal Path Finding. We devise two alternative semantic matching relations: (1) Bi-Direction, the semantic matching edges $\mathcal{E}_m$ selected by heuristic edge cutting algorithm are replaced with bi-directional edges. (2) Text2Video, the edges in $\mathcal{E}_m$ will be reversed. In the original PHGC method, each edge in $\mathcal{E}_m$ follows $V \rightarrow D$, which can be denoted as Video2Text. We conduct experiments on the EgoTV dataset, and the results are illustrated in Fig. 3.

By observing the results of each ablated model across diverse scenarios, we can see that: (1) The Bi-Direction configuration, which applies symmetric edges for semantic

matching, yields the lowest performance across all evaluation metrics. This is particularly evident in the NS and NSC metrics. We speculate this is because the lack of directional specificity in matching leads to ambiguity, causing difficulties in correctly interpreting the temporal and sequential dynamics in the multimodal data. (2) The Video2Text edges which are implemented in the original PHGC model, consistently outperform the other two setups. The superior performance indicates that aligning video features to text instructions allows for a more effective and precise task verification process, as the model can leverage the hierarchical cues from the video to better match the semantics of rules.

EgoTV				
Edge Settings	NT	NS	NSC	ABS
$\mathcal{E}_d - \{\mathcal{E} : \mathcal{V}_d \rightarrow \mathcal{V}_s\}$	57.5	69.6	85.4	82.1
$\mathcal{E}_d - \{\mathcal{E} : \mathcal{V}_d \rightarrow \mathcal{V}_r\}$	90.2	58.5	81.6	85.2
$\mathcal{E}_d$	91.7	82.4	87.1	88.0

CSV-NL				
Edge Settings	NT	NS	NO	AVG
$\mathcal{E}_d - \{\mathcal{E} : \mathcal{V}_d \rightarrow \mathcal{V}_s\}$	60.8	65.3	63.5	63.2
$\mathcal{E}_d - \{\mathcal{E} : \mathcal{V}_d \rightarrow \mathcal{V}_r\}$	64.5	61.6	60.6	62.2
$\mathcal{E}_d$	67.3	67.6	69.3	68.0

Table 4. Analysis on the edges $\mathcal{E}_d$ directed to the discriminative entity $\mathcal{V}_d$ in the DER module.

Analysis on Discriminative Entity Representation. We also devise more ablated models regarding the DER module to explore its affects on model performance. Considering the edge $\mathcal{E}_d$, which connects the implicit discriminative entities $\mathcal{V}_d$ to other vertices, we devise two ablated models: (1) $\mathcal{E}_d - \{\mathcal{E} : \mathcal{V}_d \rightarrow \mathcal{V}_s\}$ and (2) $\mathcal{E}_d - \{\mathcal{E} : \mathcal{V}_d \rightarrow \mathcal{V}_r\}$. These denote the edges connecting $\mathcal{V}_d$ to rule vertices $\mathcal{V}_r$ and to objective state vertices $\mathcal{V}_s$ respectively.

According to the results in Table 4, we reach these insights: (1) When the edges connecting the discriminative entity $\mathcal{V}_d$ to the objective state vertices $\mathcal{V}_s$ are removed, we observe a significant drop in performance across all metrics on both datasets. This demonstrates that the direct relationship between discriminative entities and the objective states is critical for accurate task verification and performance. (2) Compared to the ablated model where the edges connecting the discriminative entity $\mathcal{V}_d$ to the rule vertices $\mathcal{V}_r$ are removed, the full DER module with $\mathcal{E}_d$ consistently outperforms others across all metrics. These are because the discriminative entities are ending vertices in our procedural heterogeneous graph. It is responsible for perceiving hierarchical cues and aggregating them for final discrimination.

Analysis on Model Convergence. We also conduct more analysis on the model convergence of different ablated models, including *PHGC w/o CMRF* and *PHGC w/o DER*, to observe influence of these key components on the training process. In terms of the results illustrated in Fig. 4, we

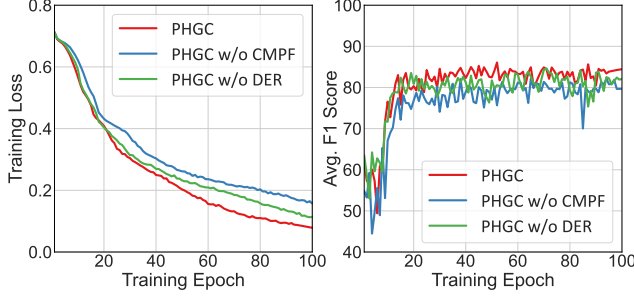


Figure 4. Analysis on model convergence regarding training loss and average performance on the EgoTV dataset.

list following observations: (1) The original PHGC model demonstrates faster and more stable convergence compared to its ablated versions. Additionally, the complete PHGC model also achieves a higher and more stable average F1 score over epochs. One probable reason is the absence of CMPF and DER modules hinders the model’s ability to align cross-modal semantic relations, resulting in slower adaptation during training. (2) The training loss decreases all the time but the performance of models tends to get decreased until several early epochs. We speculate this is due to the task verification goal can be regarded as a binary classification problem, where the random models may perform false positive results but conduct invalid verification.

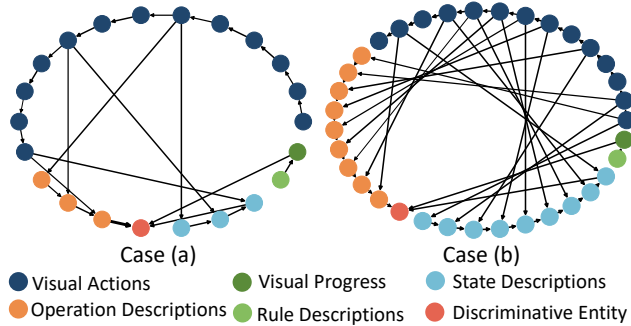


Figure 5. Procedural heterogeneous graph visualizations. Case (a) and (b) are from the EgoTV and CSV-NL datasets respectively.

Visualizations of Procedural Heterogeneous Graph. To show how our graph is constructed explicitly, we visualize the procedural heterogeneous graph of two test cases in Fig. 5. Particularly, the thickness of directed edges represent the attentive weights α in the DER module. By observing these visualization results, we can verify that our procedural heterogeneous graph is a typical directed acyclic graph, which containing vertices with hierarchical semantics and diverse edges. It represents the language-based task verification process from hierarchical views and thoroughly depicts the fine-grained cross-modal alignment. Additionally, benefit from our proposed DER module, the discriminative entity is able to aggregate the information from the whole heterogeneous graph, thus effectively integrating the most informative cues for task verification.

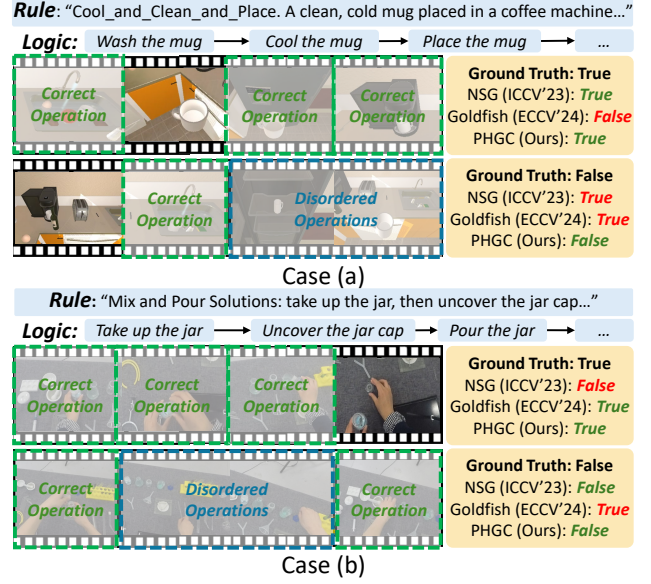


Figure 6. Qualitative analysis of the task verification on the EgoTV (Case a) and CSV-NL (Case b) datasets respectively.

Qualitative Analysis. We additionally visualize the synthetic and realistic cases from EgoTV and CSV-NL datasets to conduct a qualitative analysis of our proposed PHGC method. As illustrated in Fig. 6, we can observe that our proposed PHGC method succeeds in discriminating whether the given video fits the given language rules, while the two counterpart methods, *i.e.*, the NLETV method NSG [5] and MLLM method GoldFish [50] give incorrect results. Particularly, in Case (b), when the rules are maintained but videos are changed to different action flows, our PHGC method can alarm the operators who disobey the pre-defined rules, while the other counterpart methods fail to handle it. These results prove again that our proposed PHGC method is superior to other methods in capturing the coherent topological order and cross-modal alignment.

5. Conclusion

In this paper, we proposed a novel NLETV approach, termed Procedural Heterogeneous Graph Completion (PHGC). Additionally, we introduced the CSV-NL benchmark dataset, which enhanced real-world NLETV evaluation. Extensive experiments on both synthetic and realistic datasets consistently demonstrate the effectiveness of our proposed PHGC method. For future works, we will explore more effective NLETV method on larger realistic dataset.

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